ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C9 · LIGHT AND SOUND

Sound and light facts

★☆☆
Grade 4

Learning goals

- Recall how sound is made and travels.
- Use absorb, reflect and refract.

1 What vibrates to make the sound? [4]

Instrument	What vibrates?
dhol (drum)	the drum skin
ektara	the string
banshi (flute)	the air inside
your voice	your vocal cords

2 Absorbed, reflected or refracted? [5]

What happens	Absorbed, reflected or refracted?
you see yourself in a mirror	reflected
a black shirt gets hot in the sun	absorbed
a straw looks bent in a glass of water	refracted
thick curtains make a room quieter	absorbed
you hear an echo in a big empty hall	reflected

3 Fill in the gaps. [4]

WORD BANK

solids **higher** **louder** **vibrations** **lower** **gases**

Sounds are made by vibrations. Sound travels best through solids. A bigger vibration makes a louder sound. Faster vibrations make a higher pitch.

4 True or false? [3]

Statement	True	False	If false, correct it
1. Sound can travel through water.	✔		
2. Sound can travel through empty space.		✔	There are no particles to pass the vibration on.
3. A shorter string makes a higher pitch.	✔		

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C9 · LIGHT AND SOUND

Explaining sound and light

★★☆
Grade 4

Learning goals

- Explain pitch and loudness.
- Explain what we see.

1 **A Baul singer tightens the string of the ektara. What happens to the pitch? Why?** [2]

It gets higher, because a tighter string vibrates faster.

2 **Why can you hear a train coming sooner if you put your ear to the railway track? (Never try this!)** [2]

Sound travels faster and better through solid steel than through air.

3 **Why do recording studios have soft foam on the walls?** [2]

Soft foam absorbs sound, so there are no echoes and outside noise is muffled.

4 **A white cat and a black cat sit in the sun. Which one looks brighter, and which one gets hotter? Explain.** [2]

The white cat looks brighter because it reflects more light; the black cat gets hotter because it absorbs more light.

5 **Draw arrows to show how you see a mango on a table in a sunny room.** [2]

Seeing a mango

Arrow from the Sun to the mango, then from the mango to the eye.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

INVESTIGATE · C9 · LIGHT AND SOUND

Which material muffles sound best?

★★☆
Grade 4

Learning goals

- Plan a fair test.
- Measure loudness in decibels.
- Draw conclusions.

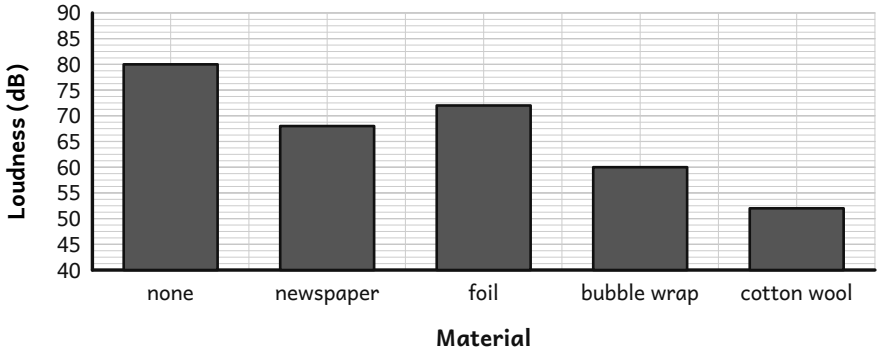
Class 4D put a phone playing a beep at the same volume inside a box. They covered the box with different materials and measured the loudness 1 metre away with a sound meter app.

1 **Complete the plan.** [3]

Change	the material covering the box
Measure	loudness (dB) at 1 m
Keep the same	the phone, the beep and its volume, the box, the distance, the same quiet room

2 **Draw a bar chart.** [3]

Material	no cover	newspaper	foil	bubble wrap	cotton wool
Loudness (dB)	80	68	72	60	52



3 **Which material muffled the sound best? Suggest why.** [2]

Cotton wool (52 dB): it is soft and thick with lots of air spaces that absorb the vibrations.

4 **Why did they measure "no cover" first?** [1]

To compare: it shows how loud the sound is without any muffling.

5 **Which material would you use to make ear defenders? Why?** [1]

Cotton wool or foam, because it absorbed the most sound.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C9 · LIGHT AND SOUND

The music of the Bauls

★★★
Grade 4

Learning goals

- Apply sound science to traditional instruments.
- Link science with culture.
- Design an instrument.

Science · Music · Culture · Design

The **Bauls** are wandering singers of Bengal. Their songs, like those of **Lalon Shah**, talk about love, humanity and living simply. In **2005**, UNESCO recognised Baul songs as a treasure of the world's culture.

Bauls play simple instruments, often made by hand. The **ektara** has one string stretched over a hollow gourd; squeezing its bamboo neck changes the tension and the pitch. The **dotara** has more strings. The **dugi** (a small drum) is beaten by hand, and the **ghungur** (ankle bells) jingle as the singer dances. The hollow bodies of these instruments make the sound louder by vibrating with the strings.

Based on information about Baul music of Bangladesh (UNESCO Intangible Cultural Heritage).

1 **Complete the table.** [6]

Instrument	What vibrates?	How do you change the pitch?
ektara	the string	squeeze the neck to change the string tension
dugi (drum)	the drum skin	tighten the skin or use a smaller drum
banshi (flute)	the air inside	cover holes to change the length of the air column

2 **Why does the ektara have a hollow gourd at the bottom?** [2]

The hollow gourd vibrates with the string and makes the sound louder.

3 **Design your own instrument from recycled materials. Draw it and explain how to make *a* high note *and a* low note.** [3]

My instrument

e.g. rubber bands of different thickness over a box: thin/tight bands give high notes; thick/loose bands give low notes.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C9 · LIGHT AND SOUND

Show what you know

★★★
Grade 4

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

A	A	P	J	P	I	T	C	H	E
B	S	N	D	D	O	S	G	J	H
S	E	R	B	P	L	W	D	H	W
O	L	C	E	L	S	I	T	T	G
R	O	S	H	F	I	O	E	P	W
B	U	J	T	O	L	G	U	J	E
U	D	E	B	A	K	E	H	N	O
T	M	I	R	R	O	R	C	T	D
V	I	B	R	A	T	E	K	T	M
U	N	T	M	H	U	I	W	S	E

absorb echo light loud mirror pitch reflect sound vibrate

2 Which material is best at absorbing sound? [1]

○ glass
○ metal
● **thick cloth**
○ tiles

3 A pencil in water looks bent because light is... [1]

○ absorbed
○ reflected
● **refracted**
○ stopped

4 Draw two waves: a loud low sound and a soft high sound. [2]

loud, low
tall waves, far apart

soft, high
short waves, close together

5 Challenge. Astronauts on the Moon talk to each other by radio, even when standing side by side. Explain why. [2]

There is no air on the Moon, so there are no particles to carry sound vibrations; radio signals can travel through empty space.